

CO4 ASSESSMENT TOOL-3(Comparative Analysis)

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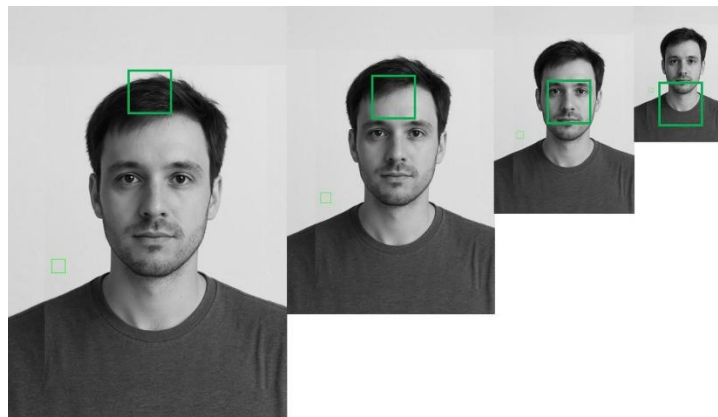
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1. Object Detection Performance Comparison

Two object detection models, Viola–Jones (Model A) and YOLO (Model B), are evaluated using True Positives (TP), False Positives (FP), and False Negatives (FN).

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A – Viola–Jones	80	20	30
Model B – YOLO	90	15	20



(a) Precision and Recall

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model A – Viola–Jones:

- Precision = $80 / (80 + 20) = 0.80 = 80\%$

- $\text{Recall} = 80 / (80 + 30) = 0.727 = 72.73\%$

Model B – YOLO:

- $\text{Precision} = 90 / (90 + 15) = 0.857 = 85.71\%$
- $\text{Recall} = 90 / (90 + 20) = 0.818 = 81.82\%$

Comparison

Performance Measure	Viola–Jones	YOLO
Precision	80%	85.71%
Recall	72.73%	81.82%

(b) Overall Performance and Justification

From the calculated values, **YOLO** performs better than Viola–Jones in both precision and recall.

The precision of YOLO is 85.71%, compared with 80% for Viola–Jones. This means YOLO produces fewer incorrect object detections. YOLO also has a recall of 81.82%, which is higher than the 72.73% recall of Viola–Jones. Therefore, YOLO is able to detect a larger proportion of the actual objects present.

Another important observation is that YOLO has 90 true positives, while Viola–Jones has only 80 true positives. YOLO also has fewer false positives (15 compared with 20) and fewer false negatives (20 compared with 30).

Thus, YOLO provides a better balance between correctly detecting objects and avoiding incorrect detections.

2.Motion Estimation Error Analysis

Motion estimation is an important technique in computer vision and video processing. It determines how an object or a pixel moves from one video frame to the next. The movement is generally represented using a displacement or motion vector.

In this problem, two motion estimation techniques are evaluated based on their displacement errors over three consecutive frames.

Given Data

Frame	Method A Error	Method B Error
Frame 1	2 pixels	3 pixels
Frame 2	3 pixels	5 pixels
Frame 3	2 pixels	4 pixels

The displacement error represents the difference between the actual motion and the motion estimated by the technique. A smaller error indicates better accuracy.

(a) Calculate the Average Displacement Error

The average displacement error is calculated using:

Average Error = Sum of displacement errors / Number of frames

Method A

The errors for Method A are: 2,3,2

$$\text{Average Error} = (2 + 3 + 2) / 3$$

$$\text{Average Error} = 7 / 3$$

$$\text{Average Error} = 2.33 \text{ pixels}$$

Therefore, the average displacement error of Method A is 2.33 pixels.

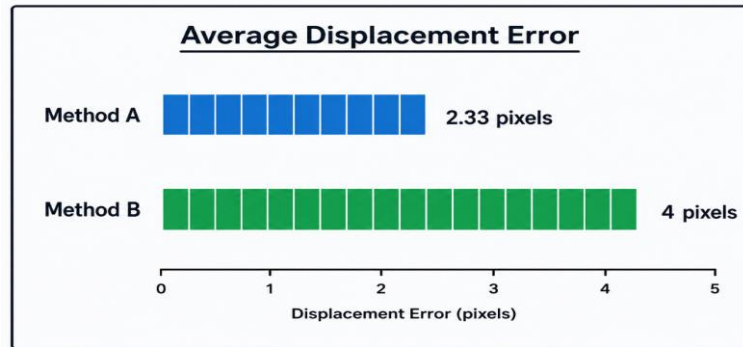
For Method B:

$$\text{Average Error} = (3 + 5 + 4) / 3$$

$$\text{Average Error} = 12 / 3$$

$$\text{Average Error} = 4 \text{ pixels}$$

Therefore, the average displacement error of Method B is 4 pixels.



Comparison Table

Parameter	Method A	Method B
Frame 1 Error	2 pixels	3 pixels
Frame 2 Error	3 pixels	5 pixels
Frame 3 Error	2 pixels	4 pixels
Total Error	7 pixels	12 pixels
Average Error	2.33 pixels	4 pixels
Accuracy	Higher	Lower
Overall Performance	Better	Poorer

Comparison: Method A has a lower average displacement error of 2.33 pixels, whereas Method B has an average error of 4 pixels. Therefore, Method A provides more accurate motion estimation because lower displacement error indicates better accuracy.

Justification Based on Error Values

In motion estimation, the objective is to estimate the movement of an object as closely as possible to its actual movement.

A smaller displacement error means that the estimated motion vector is closer to the actual motion vector. Therefore, the technique with the smaller average error provides more accurate motion estimation.

For Method A:

- Minimum error = 2 pixels

- Maximum error = 3 pixels
- Average error = 2.33 pixels

For Method B:

- Minimum error = 3 pixels
- Maximum error = 5 pixels
- Average error = 4 pixels

Method A not only has a lower average error, but it also has a smaller maximum error. This means its estimates are more consistent across the three frames.

Difference in Average Error

Thus, Method A has an average error that is 1.67 pixels lower than Method B.

The percentage reduction in average error is approximately:

So, Method A's average displacement error is approximately 41.75% lower than Method B's

CONCLUSION

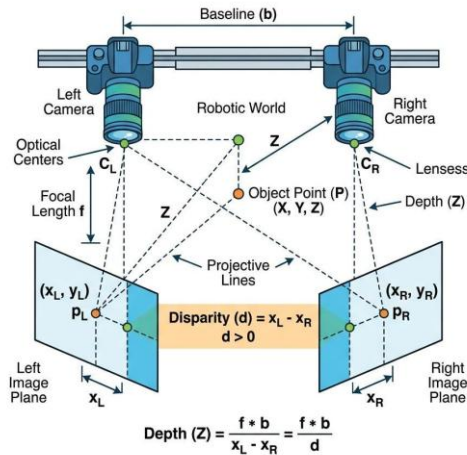
Method A is more accurate than Method B because its average displacement error is only 2.33 pixels, while Method B has an average error of 4 pixels. Since lower displacement error indicates better accuracy, Method A is the preferred motion estimation technique.

3.Depth Estimation Comparison

Given Data

Object	Actual Depth	Stereo Vision	SfM
Object 1	2.0 m	2.0 m	2.5 m
Object 2	3.0 m	3.0 m	3.8 m
Object 3	4.0 m	4.0 m	5.0 m

Stereo Vision in Robotics



(a) Absolute Error for Each Object

Formula:

Absolute Error = |Estimated Depth – Actual Depth|

Stereo Vision:

- Object 1: $|2.0 - 2.0| = 0.0 \text{ m}$
- Object 2: $|3.0 - 3.0| = 0.0 \text{ m}$
- Object 3: $|4.0 - 4.0| = 0.0 \text{ m}$

SfM:

- Object 1: $|2.5 - 2.0| = 0.5 \text{ m}$
- Object 2: $|3.8 - 3.0| = 0.8 \text{ m}$
- Object 3: $|5.0 - 4.0| = 1.0 \text{ m}$

(b) Average Absolute Error

Stereo Vision:

Average Error = $(0.0 + 0.0 + 0.0) / 3$
= 0.0 m

SfM:

Average Error = $(0.5 + 0.8 + 1.0) / 3$

$$= 2.3 / 3$$

$$= \mathbf{0.77\ m}$$

(c) Comparison

Method	Absolute Errors	Average Absolute Error
Stereo Vision	0.0, 0.0, 0.0 m	0.0 m
SfM	0.5, 0.8, 1.0 m	0.77 m

Conclusion: Stereo Vision provides more accurate depth estimation because its average absolute error is 0.0 m, while SfM has an average error of 0.77 m. A lower error indicates that the estimated depth is closer to the actual depth.